

A One-Stop Shop for the Custom-Made LTE Stack



Priya Ravindran is a technical journalist at EFM

In today's world, no one wants to be stuck at one place while using Internet-connected devices or carry around bulky wires along with the equipment they use. Wireless technology is ruling the world and there is a growing demand for quality service at higher speeds.

This fact gave Mymo Wireless the insight to develop Long Term Evolution (LTE) standard for wireless communication. The need for developing indigenous technology that could create a benchmark in the Indian telecommunication industry as well as globally, provided the impetus to come up with a 4G LTE user equipment (UE) modem.

The product

'A one-stop shop for the entire custom-made LTE stack' is the developers' proposition for the 4G modem. Developed using indigenous high-performance based systems and in-house products, the design uses a unique baseband receiver and other such algorithms, with optimised implementations. Having taken seven years from concept to completion, the modem is customisable for different applications in the domains of research and development, telecom, enterprise, military and Internet of Things (IoT)/machine-to-machine (M2M) communications, with upgradation in silicon, field programmable gate arrays (FPGAs) and digital signal processing (DSP). It also has many patents to its credit.

The design technology

The UE modem is built along LTE, also called 4G LTE, a standard for wireless communication of high-speed data for mobile phones and data terminals. Based on Global System for Mobile Communications

(GSM)/Enhanced Data rates for GSM Evolution (EDGE) and Universal Mobile Telecommunications System (UMTS)/High-Speed Packet Access (HSPA) network technologies, LTE technology aims to increase capacity and speed using a different radio interface, together with core network improvements. Advanced DSP techniques and modulations have been used to achieve this and the network architecture is Internet Protocol (IP) based to reduce transfer latency. It operates on the LTE spectrum.

Their brainchild becomes the brain behind. Mymo developed an LTE UE ultra software radio (USR) called MWx000 (x denotes the generation). This radio provisions accessing baseband in real-time radio frequency (RF), flexibility in its coding and validation that is not possible even with DSP and FPGAs. This extends the capabilities of the radio and allows researchers to explore new algorithms.

MW2000 is a software defined radio (SDR), a radio communication system that is implemented completely by software configurations. The SDR is classified into tiers by Wireless Innovation Forum (WINNF) depending on the features offered by it, and MW2000 falls under Tier 4 category. This is the USR offering complete programmability and supporting a broad range of functions and frequencies at the same time. The functioning is very similar to that of a multifunction phone.

Overcoming the challenges

The major challenge faced by the team was financial constraints. Initially, Mymo sought funds for the investment from the Indian ecosystem, but that did not work out and they say that it was a challenge for them to sustain the company without external funding or investment. Continuously on the look-out for opportunities to generate revenue, they started licensing their technology to Tier 1 industries in the USA. As

The 4G LTE UE modem

